

ARSHVIR

Bengaluru, India | +91 7528953747 | arshvir882@gmail.com | [LinkedIn \(linkedin.com/in/arshvir-995a91203/\)](https://www.linkedin.com/in/arshvir-995a91203/) | [GitHub \(github.com/avarshvir\)](https://github.com/avarshvir)

Professional Experience

Netscout | Bengaluru, India

AI Engineering Intern | April 2026 – Present

- Developed an AI-powered DOM Locator Service to identify and retrieve web elements for browser automation.
- Built embedding-based element matching and retrieval systems to improve locator accuracy across dynamic web applications.
- Contributing to the integration of the DOM Locator Service with Netscout products such as NG1 and OCI.
- Evaluating embedding models and proposing solutions, including IBM Granite Embedding, to improve element search and retrieval performance.
- Currently working in Computer Vision Project for the Organization.

Skills

Programming Languages: C, C++, Python

Databases: MySQL, Vector Databases (ChromaDB)

AI/Machine Learning: Scikit-Learn, TensorFlow, PyTorch, NLP, YOLO, Transformers, Diffusion Models, RAG

Software Engineering: DSA, Object Oriented Programming, Git, GitHub, Linux, Software Development, JIRA

Applied AI & Tools: Streamlit, Gradio, OpenCV, Fast API, Ollama (Local LLM Inference), Web Scraping, Google Colab, Render

Projects

Visio AI (Machine Learning and Data Science Platform) [\[Link\]](#)

- Developed an end-to-end ML and data science platform enabling dataset upload, automated data cleaning, EDA, and visualization for supervised and unsupervised learning workflows.
- Implemented training pipelines for regression, classification, and clustering algorithms, with intelligent algorithm suggestions based on target variables.
- Enabled model export functionality allowing users to download trained models for reuse in external applications.
- Integrated OpenRouter API for visual LLM to find hidden patterns of images, extending platform capabilities beyond tabular data.
- Built a no-code, user-friendly interface using Streamlit, enabling non-technical users to perform end-to-end ML workflows.

PPE Object Detection System [\[Link\]](#)

- Developed a deep learning-based PPE object detection system to automatically identify safety equipment (Helmet, Vest, Gloves, Boots) in industrial environments.
- Fine-tuned a pretrained YOLOv8 model using transfer learning on a custom Roboflow dataset and trained it on Google Colab.
- Evaluated model performance using IoU-based metrics including Precision (89%), Recall (89%), mAP@50 (91%), and mAP@50-95.
- Built an AI-powered web application using Gradio and OpenCV for image and real-time camera-based PPE compliance detection.

Fission AI (Multimodal Generative AI System) [\[Link\]](#)

- Built a multimodal generative AI platform supporting text-to-image, text-to-video, text-to-text, and text-to-deep-research functionalities.
- Integrated Stable Diffusion for image generation and BLIP for image captioning, followed by narrated video creation by stitching generated images and adding speech using gTTS.
- Implemented Local LLMs (Gemma via Ollama) for privacy-preserving text generation and summarization.
- Designed an intelligent research agent that extracts keywords, performs web searches via DuckDuckGo API, and generates reports.
- Developed a Streamlit frontend connected to a Colab backend via ngrok, demonstrating end-to-end system integration.

NGIBS – Next Gen. Intelligent Browsing System [\[Link\]](#)

- Developed NGIBS, a privacy-focused AI research and browsing system that runs fully locally on user hardware.
- Implemented multiple AI intelligence modes including Quick Search, Live Search, Deep Search, and Context Mode to support offline knowledge retrieval, reasoning, and memory-aware conversations.
- Designed a modular architecture allowing users to download and integrate any compatible LLM for local inference.
- Integrated Ollama as the backend engine to manage local model execution and inference.

Education

Sardar Beant Singh State University | Gurdaspur, India

B. Tech in Computer Science and Engineering | 2021 – 2025

CGPA: 8.12

Govt. Sen. Sec. School | Tanda, Punjab

12th Non. Medical - 87 % | 2021

Certificates

- AI/ML for Geodata Analysis, ISRO-IIRS (Aug 2024)
- Machine Learning Training, Internshala (July 2024)